

AE-376: Radiation of [Ru(CO)(OH)₂(PNP{tBu})] (AE-310) 365 nm and white, 5 mM Na₂CO₃

Date: 2024-11-13

Tags: Dihydroxo Radiation PNP
P(tBu)N(py)P(tBu) [Ru(CO)(OH)₂(PNP)]
NMR AE 1H 31P

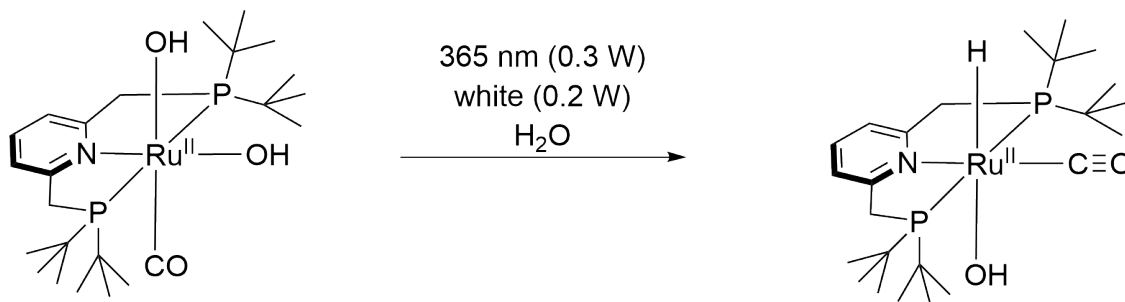
Category: Two-photon

Status: Done

Created by: Alexander Eith

Reaction scheme/sample structure

Radiation of Dihydroxo



Chemical Formula: C₂₄H₄₅NO₃P₂Ru
Molecular Weight: 558,64552
[Ru(CO)(OH)₂(PNP^{tBu})]

Chemical Formula: C₂₄H₄₅NO₂P₂Ru
Molecular Weight: 542,64652
[Ru(CO)H(OH)(PNP^{tBu})]

Literature/reference experiments

Literature	Procedure after https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiment	Two-photon - AE-357: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 1000 mM KOH Two-photon - AE-358: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 50 mM KOH Two-photon - AE-362: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 5 mM KOH I Two-photon - AE-363: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 1 mM KOH Two-photon - AE-371: Radiation of [Ru(CO)(OH) ₂ (PNP{tBu})] (AE-310) 365 nm and white, 0.1 mM KOH

Reagents

Name	CAS/Experiment number	Amount [μmol]	Equivalents	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
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[Ru(CO)(OH)2(PNP{tBu})]	AE-310: Synthesis of [Ru(CO)(OH)2(PNPtBu)] using Ag2O, 5 h (AE-310-2)	/	/	2.50	/	558.65	/
Na2CO3 (0.1 M)	Two-photon - AE-375: Stock solutions of Na2PO4 and Na2CO3 in H2O and KOH in D2O (AE-375-2)	/	/	/	/	/	0.045
Milli-Q water	KRA-104: Degassing of Milli-Q water	/	/	/	/	/	0.85

Irradiation Parameters

Power measured using [Power Meter] 843-R-USB + 919P-020-12 unless specified otherwise.

	Light Source Name	Wavelength [nm]	Power Setting [mW]
First light source	Light Source - LCS-0365-76-22	365 nm	300
Second light source	Light Source - LCS-6500-65-22	white	200

Used beam combiner [Name or None]	Beam Combiner - LCS-BC25-0409
Irradiation distance [cm]	9.5
Thermostat temperature [°C]	25
Irradiation duration [h]	3

Procedure/observations

All steps were carried out using [Protocol] Schlenk Technique. All steps were carried out in the dark, unless other specified.

Date	Time	Step	Observations
12.11		A 10 mL flask was prepared	
	15:00	To the flask containing the remaining [Ru] from AE-310-2 water was added using a 1 mL syringe	
		To that flask Na2CO3 (0.1 M) was added using a 50 µL Hamilton syringe	
		The obtained mixture was transferred into the prepared 10 mL flask	
	15:12 - 15:22	The obtained mixture was stirred at 400 rpm.	before filt.jpg

	15:29	From the obtained solution an NMR sample was prepared according to Protocol - Preparation of NMR Sample using a D2O/H3PO4 capillary. (AE-376-1) During the transfer the sample was filtered through a syringe filter (PA, pore diameter = 0.2 µm)	before irrad.jpg
14.11	13:15	The power of the LEDs 365 nm: 300 mW and white: 200 mW were adjusted according to [Protocol] Irradiation setup	power setup.jpg
		The NMR-tube was placed in the setup according to [Protocol] Irradiation setup	irrad setup 1.jpg irrad setup 2.jpg
	10:05 - 13:05	The J.Young NMR tube was irradiated for 3:00 h	
	16:50	An NMR was measured (AE-376-2)	AE-376-2.jpg

Analysis

Date	Time	Sample name	Analysis Method	Used Device	Solvent	Raw Data	Processed Data	Comparative Data	Interpretation
13.11	01:01	AE-376-1	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-376-1.zip	AE-376-2_10.nmrium	/	As expected
15.11	15:11	AE-376-2	31P {1H}quant, 1Hzgespg	IAAC 400 MHz I	H2O	AE-376-2.zip	AE-376-2_10.nmrium	/	As expected

Results

Sample	Yield (hydride species, cis) [%]	Yield (hydride species, trans) [%]	Yield (dihydroxo) [%]	Yield (main side product) [%]	Yield (soluable side products) [%]	Yield (other side products) [%]	Data
AE-376-2	9	0	73	20	4	-6	AE-376-2_Yield_Determination.xlsx

Linked experiments

- [AEI-022: Radiation of \[Ru\(CO\)\(OH\)2\(PNN\)\], NO2-route, 365 nm and white LED](#)

- [AEI-026: Capillary production](#)

- AEI-064: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O

- AEI-067: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, crude, 365 nm and white

- AEI-071: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purer, 365 nm and white

- AE-105: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O

- AE-110: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white

- AE-119: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 52 mM KOH, 365 nm and white

- AE-140: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white, longer duration I

- AE-145: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP})]$ with 2,6-bis(P,P-di(tertbutyl)phosphinomethyl) using Ag_2O

- AE-158: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$, purified, 365 nm and white

- KRA-092: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$, 365 nm and white

- KRA-104: Degassing of Milli-Q water

- AE-238: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

- AE-243: 1 M Stocksolution of KOH in water

- AE-246: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-238, 97 % pure) 365 nm and white

- AE-310: Synthesis of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNPtBu})]$ using Ag_2O , 5 h

Prep work - AE-361: 5 mM Stocksolution of KOH in water

Two-photon - AE-348: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, water, longer irradiation

Two-photon - AE-349: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH, shorter irradiation

Two-photon - AE-357: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1000 mM KOH

Two-photon - AE-358: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 50 mM KOH

Two-photon - AE-362: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 5 mM KOH

Two-photon - AE-363: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 1 mM KOH

Two-photon - AE-371: Radiation of $[\text{Ru}(\text{CO})(\text{OH})_2(\text{PNP}\{\text{tBu}\})]$ (AE-310) 365 nm and white, 0.1 mM KOH

Two-photon - [AE-375: Stock solutions of Na₂PO₄ and Na₂CO₃ in H₂O and KOH in D₂O](#)

Linked resources

Beam Combiner - [Beam Combiner LCS-BC25-0409 -1](#)

Light Source - [UHP LED 365 nm-1](#)

Light Source - [UHP LED 6500 K \(white\)-1](#)

Protocol - [Preparation of NMR Sample](#)

Protocol - [Schlenk Technique](#)

Protocol - [Irradiation setup](#)

Attached files

AE-376-2_Yield_Determination.xlsx

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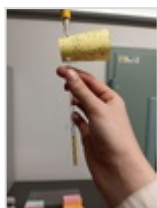
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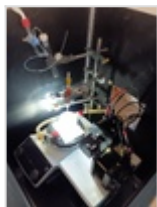
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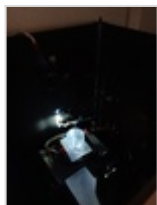
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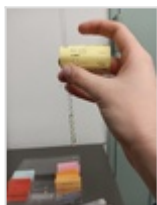
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before-irrad.jpg

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before-filt.jpg

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Unique eLabID: 20241113-2a735bf3b3b1201264254ead01c410db792fa55a
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1496>